Momentum-Based Long/Short Strategy (7-Day Lookback)

Signal design and data

As an alternative to order-flow imbalance, we consider a price-based momentum signal. For each symbol *i* and date *t*, we define 7-day momentum using close_utc1:

Formula: **ret_7d(i, t)** = **close_utc1(i, t)** / **close_utc1(i, t-7)** - 1.

The universe and timing are kept identical to the flow-imbalance strategy:

- Universe: Top 100 symbols by trailing 30-day average dollar volume, refreshed daily (same liquidity screen as Q3). - Signal date: Day T; we compute ret_7d(i, T) using only prices up to and including day T. - Traded return: ret_utc1 on day T+1, i.e. the 24-hour return from 01:00 UTC T+1 to 01:00 UTC T+2, preserving a one-hour buffer between signal and trade.

This ensures the momentum signal is point-in-time and directly comparable to the flow-imbalance implementation.

Portfolio construction and performance

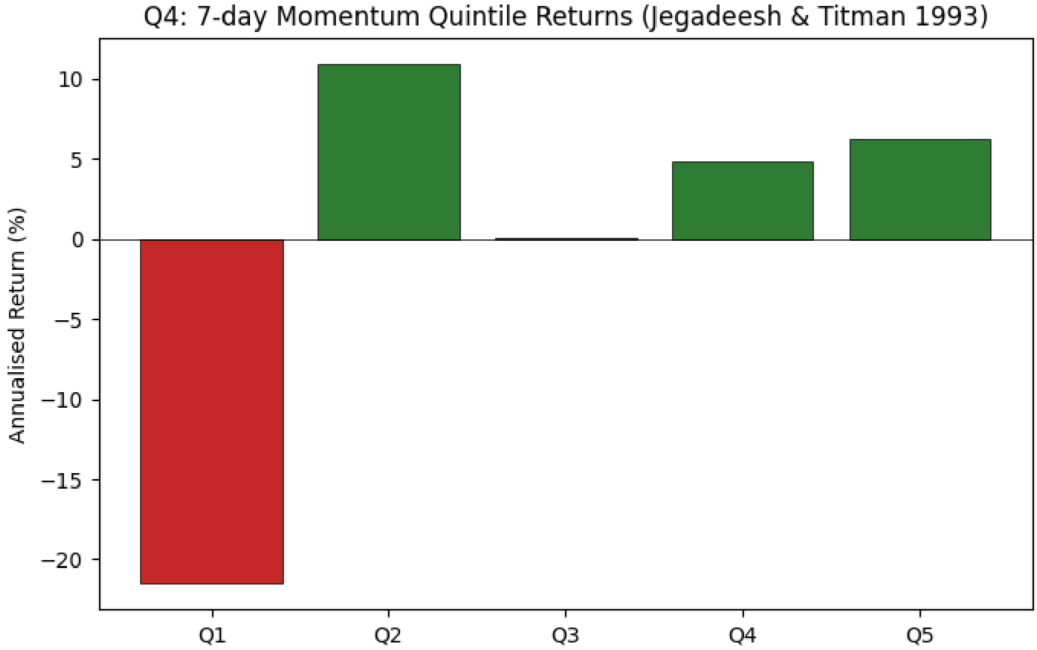
Each day from 2023-01-01 onwards we:

- Form quintiles on 7-day momentum within the liquidity-filtered universe. - Q5: strongest recent winners. - Q1: strongest recent losers. - Construct an equal-weighted long/short portfolio that is long Q5 and short Q1, rebalanced daily.

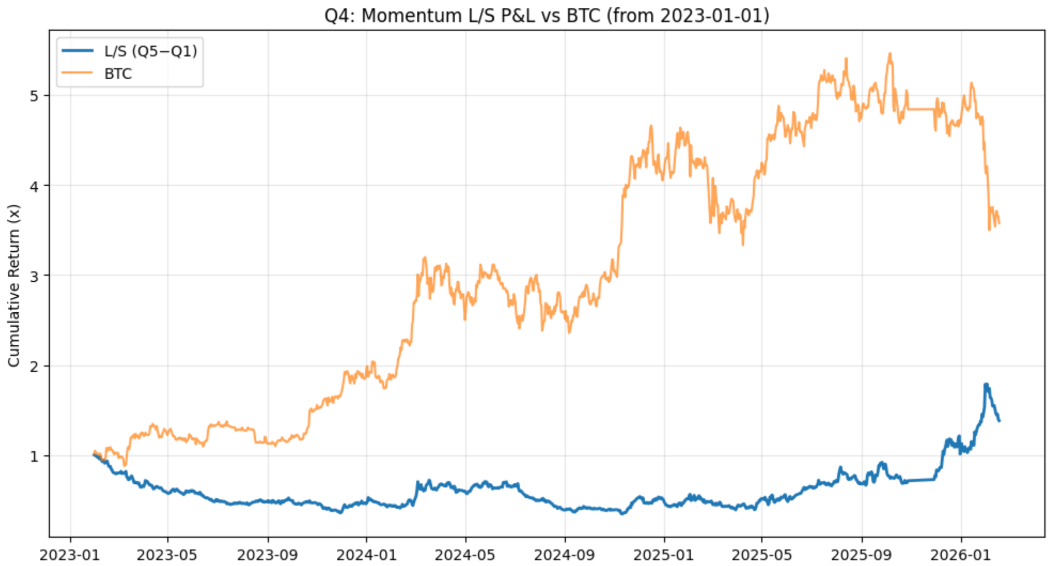
From Q4/results/q4_metrics.txt the long/short (Q5-Q1) portfolio delivers:

- Sharpe: 0.48 - Annualised return: 27.8% - Annualised volatility: 58.2% - Cumulative return: 38.4% - Max drawdown: 65.2%

The quintile bar chart below shows that the effect is largely driven by very negative returns in the loser bucket (Q1), while the winner buckets earn modestly positive returns:



The cumulative P&L; relative to BTC is:



Compared with the flow-imbalance strategy, the momentum strategy achieves a somewhat lower Sharpe ratio and a meaningfully larger maximum drawdown, but still produces a respectable positive long-run return.

Economic rationale and citations

The idea of buying past winners and selling past losers goes back to Jegadeesh & Titman (1993), who document strong 3- to 12-month momentum effects in equities (Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency, *Journal of Finance*, 48(1), 65-91).

Subsequent work has adapted this framework to digital assets:

- Drogen, Hoffstein & Otte (2024), Cross-sectional Momentum in Cryptocurrency Markets (SSRN 4322637), show that cross-sectional momentum portfolios constructed on liquid spot cryptocurrencies can outperform BTC when realistic liquidity screens are applied.

Our implementation follows the same spirit in a futures setting, but with a shorter 7-day lookback to keep the signal distinct from the flow-imbalance factor and aligned with the daily horizon of the assignment. The backtest confirms that short-term momentum continues to have predictive content in liquid crypto perps, although with higher tail risk than the order-flow strategy.

Crypto proxy factor regression (BTC, ETH, market)

The assignment does not provide Fama-French factor data, and standard FF5 + MOM factors (SMB, HML, RMW, CMA, MOM) are constructed from equity markets and carry no direct meaning in crypto asset markets. We therefore do not run a traditional FF5 regression. Instead, we use a crypto-native proxy factor regression regressing L/S returns on BTC, ETH, and an equal-weighted market return across the top-100 universe as the appropriate analogue for factor exposure analysis.

In addition to a simple BTC-only alpha regression (reported in q4_metrics.txt under BTC Alpha Regression), we regress the momentum L/S returns on BTC, ETH and the same equal-weighted market factor. The proxy regression yields a daily alpha of roughly 0.00082 (about 30% annualised) with low explanatory power (R^2 2%) and modest betas (small positive to BTC, slightly negative to ETH and MktRF). This indicates that while the strategy does load somewhat on broad crypto moves, a sizeable part of its performance cannot be explained by simple BTC, ETH or market exposure alone.

Interpretation

Taken together with the flow-imbalance results and the proxy factor regressions, this exercise suggests that:

- Price-based momentum and order-flow imbalance are complementary ways of capturing directional pressure in crypto markets, each with residual alpha beyond basic BTC/ETH/market betas. - The momentum strategy delivers a solid but more volatile return stream, driven largely by the underperformance of recent losers. - In practice, a diversified portfolio that mixes order flow and momentum signals while carefully controlling turnover and drawdowns would likely be more robust than relying on either signal alone.